FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Intestinal Worms

Tutorial | Week 2 · Diarrhoea, fluids, HIV & anaemia | Paediatrics | Dr Aliku

Soil-transmitted helminths (roundworm, hookworm, whipworm, Strongyloides), pinworm, tapeworms and schistosomes infect a large share of Ugandan children, causing anaemia, stunting, poor school performance and occasionally surgical emergencies; single-dose albendazole or mebendazole every 6 months, praziquantel for schistosomiasis and tapeworms, and ivermectin for Strongyloides, plus sanitation, shoes and safe water, control them.

Must know before the ward round

1. Routine deworming (UCG 2023): every 6 months from 1 year. Albendazole 200 mg (12–23 months) or 400 mg (2 years and over) single dose; or mebendazole 250 mg (1–2 years) or 500 mg (over 2 years) single dose.
2. WHO 2017 preventive chemotherapy for soil-transmitted helminths: once a year where prevalence is 20% or more, twice a year where it is 50% or more, for preschool and school-age children, adolescent girls and women of reproductive age (not in the first trimester).
3. Hookworm (Necator, Ancylostoma) enters through bare skin and sucks blood: it causes iron-deficiency anaemia and hypoalbuminaemia. Treat the worm, then give iron (elemental iron about 3 mg/kg/day for 3 months).
4. Ascaris larvae pass through the lungs (Löffler syndrome: cough, wheeze, eosinophilia); heavy adult loads cause intestinal obstruction, biliary colic, pancreatitis and appendicitis, especially in children aged 1–5 years.
5. Trichuris (whipworm) lives in the caecum and colon: heavy infection gives bloody mucoid diarrhoea, rectal prolapse, anaemia and finger clubbing; single-dose albendazole cures it poorly, so give albendazole 400 mg daily for 3 days.
6. Strongyloides is the odd one out: it multiplies inside the host (autoinfection), is missed by routine stool microscopy and by albendazole mass treatment, and steroids or HTLV-1 can trigger fatal hyperinfection. Treat with ivermectin 200 µg/kg.
7. Screen for or treat Strongyloides before starting high-dose steroids (e.g. nephrotic syndrome) in a child from an endemic area.
8. Pinworm (Enterobius): night-time perianal itch; diagnose with the sticky-tape test in the morning; albendazole single dose repeated after 2 weeks; treat the whole household.
9. Schistosomiasis: praziquantel 40 mg/kg single dose (UCG 2023; WHO 2022 annual preventive chemotherapy from age 2 where prevalence is 10% or more). S. mansoni (Lake Victoria, Albert, Kyoga, the Nile) causes periportal fibrosis, splenomegaly and variceal bleeding; S. haematobium causes terminal haematuria.
10. Tapeworm (Taenia): praziquantel 5–10 mg/kg single dose (UCG) or niclosamide by age; Hymenolepis nana: praziquantel 25 mg/kg. T. solium eggs cause neurocysticercosis, a common cause of epilepsy; never give antiparasitic treatment for cysticercosis without imaging.
11. Albendazole and mebendazole are not given routinely under 12 months; ivermectin is avoided under 15 kg; praziquantel is now given from 2 years, and a paediatric formulation (arpraziquantel) exists for younger children.
12. Eosinophilia means tissue migration (Ascaris and hookworm larvae, Strongyloides, schistosomes, Toxocara), not adult worms sitting in the gut lumen.

1 Definition
2 Epidemiology
3 Aetiology and Risk Factors
4 Pathophysiology
5 Classification / Staging
6 Clinical Features
7 Differential Diagnoses
8 Clinical Workup
9 Management
10 Complications
11 Prognosis
12 Prevention
13 Special Populations
14 Key Points
15 References

1. Definition

Intestinal worms (helminths) are parasitic worms that live in the human gut or, for schistosomes, in the veins around the gut and bladder. The big groups are:

- the **soil-transmitted helminths (STH)**: roundworm (*Ascaris*), whipworm (*Trichuris*), hookworms (*Necator*, *Ancylostoma*) and *Strongyloides*;
- **pinworm** (*Enterobius*);
- **tapeworms** (*Taenia*, *Hymenolepis*);
- **schistosomes** (bilharzia).

Most infections are light and silent. The harm comes from **heavy, repeated infection in growing children**: **iron-deficiency anaemia** (hookworm, whipworm, schistosomes), **malnutrition and stunting**, poor school performance, and sometimes **surgical emergencies** (a ball of roundworms blocking the bowel, worms in the bile duct) or a **fatal Strongyloides hyperinfection** in a child on steroids.

Group	Worm (common name)	How it gets in	Where the adult lives	Size and lifespan
STH	*Ascaris lumbricoides* (roundworm)	Swallowed **eggs** (soil on hands, unwashed vegetables)	Small intestine	15–35 cm; 1–2 years
STH	*Trichuris trichiura* (whipworm)	Swallowed eggs	**Caecum and colon**	3–5 cm; 1–3 years
STH	*Necator americanus*, *Ancylostoma duodenale* (hookworms)	**Larvae penetrate bare skin** (*Ancylostoma* also swallowed)	Small intestine, attached to the mucosa	about 1 cm; years
STH	*Strongyloides stercoralis* (threadworm in some texts)	Larvae penetrate skin; **autoinfection** in the host	Small-intestinal mucosa	2 mm; **lifelong** without treatment
Pinworm	*Enterobius vermicularis* (pinworm/threadworm)	Swallowed eggs (fingers after scratching, bedding)	Caecum; females lay eggs on the **perianal skin at night**	about 1 cm; weeks
Tapeworms	*Taenia saginata* (beef), *Taenia solium* (pork)	Undercooked meat containing **cysticerci**	Small intestine	metres; years
Tapeworms	*T. solium* eggs → **cysticercosis**	Swallowed **eggs** (faecal–oral, from a human carrier)	Larval cysts in **brain**, muscle, eye	
Tapeworms	*Hymenolepis nana* (dwarf tapeworm)	Swallowed eggs; autoinfection	Small intestine	2–4 cm; common in young children
Trematodes	*Schistosoma mansoni* (intestinal), *S. haematobium* (urogenital)	**Cercariae penetrate skin** in fresh water with snails	**Mesenteric veins** (*mansoni*), **vesical plexus** (*haematobium*)	1–2 cm; 3–10 years

CLINICAL PEARL

The one thing never to forget: **deworming is cheap and safe: albendazole 400 mg (200 mg under 2 years) every 6 months from 1 year.** But treating the child is only half the job. Without **latrines, handwashing, shoes and safe water**, reinfection returns within months.

2. Epidemiology

- **Global (WHO):** about **1.5 billion people** (24% of the world) have STH infections, with the highest prevalence in sub-Saharan Africa; over 600 million have *Strongyloides*. At least **254 million** people needed preventive treatment for schistosomiasis in 2024, about **94%** of them in Africa.
- **Uganda:**
  - STH infections are common wherever sanitation is poor; **hookworm** is the most widespread STH in many districts. Prevalence has fallen with twice-yearly deworming but rebounds where latrine coverage is low.
  - ***S. mansoni*** is highly endemic along the shores of **Lake Victoria, Lake Albert, Lake Kyoga and the Albert Nile** (fishing communities, e.g. Mayuge, Namayingo, Buliisa, Pakwach). Children who swim, fish, wash and fetch water are infected repeatedly. ***S. haematobium*** is found in a few foci in the east and north.
  - ***T. solium*** cysticercosis follows pig-keeping where pigs roam free and latrines are few; **neurocysticercosis** is an important cause of epilepsy in Ugandan children.
  - ***Strongyloides*** is under-recognised because routine wet preps miss it.

UGANDA CONTEXT

Deworming platforms in Uganda: Integrated Child Health Days (Child Days Plus) in April and October give **albendazole and vitamin A** to children from 1 year (vitamin A from 6 months); **school-based deworming** gives albendazole, plus **praziquantel** in schistosomiasis-endemic districts (the national NTD control programme); **ANC** gives a single dose of **mebendazole or albendazole in the 2nd trimester**; **IMCI** (UCG 2023) says give albendazole or mebendazole at any sick-child visit if the child is over 1 year and has not been dewormed in the last 6 months. Stool microscopy (wet prep) is available at most HC IIIs; Kato-Katz and urine filtration are mainly used in surveys.

3. Aetiology and Risk Factors

Routes of infection (the basis of prevention)

Route	Worms	Breaks the cycle
Swallowed eggs (faecal–oral, soil on hands, unwashed vegetables, bedding)	*Ascaris*, *Trichuris*, *Enterobius*, *T. solium* eggs (cysticercosis), *H. nana*	Latrines, handwashing, cut nails, wash vegetables
Larvae through bare skin	Hookworm, *Strongyloides*	Shoes, latrines
Cercariae through skin in fresh water	*S. mansoni*, *S. haematobium*	Avoid lake and slow-river water contact; safe water
Undercooked meat	*T. saginata* (beef), *T. solium* (pork)	Cook meat thoroughly; confine pigs
Autoinfection	*Strongyloides*, *H. nana*, *Enterobius* (re-swallowed eggs)	Treat fully; household treatment for pinworm

Who is at risk?

- **Preschool and school-age children** (play in soil, put hands in mouths).
- Children who walk **barefoot**, live without a **latrine**, or **swim, fish, wash or fetch water** in lakes and slow rivers.
- Communities where **pigs roam free** and pork is eaten undercooked.
- Children with **HIV** (*Strongyloides*) and **malnutrition**.
- **Strongyloides hyperinfection**: children on **steroids**, with **HTLV-1**, malnutrition or transplant.

4. Pathophysiology

Life cycles that explain the symptoms

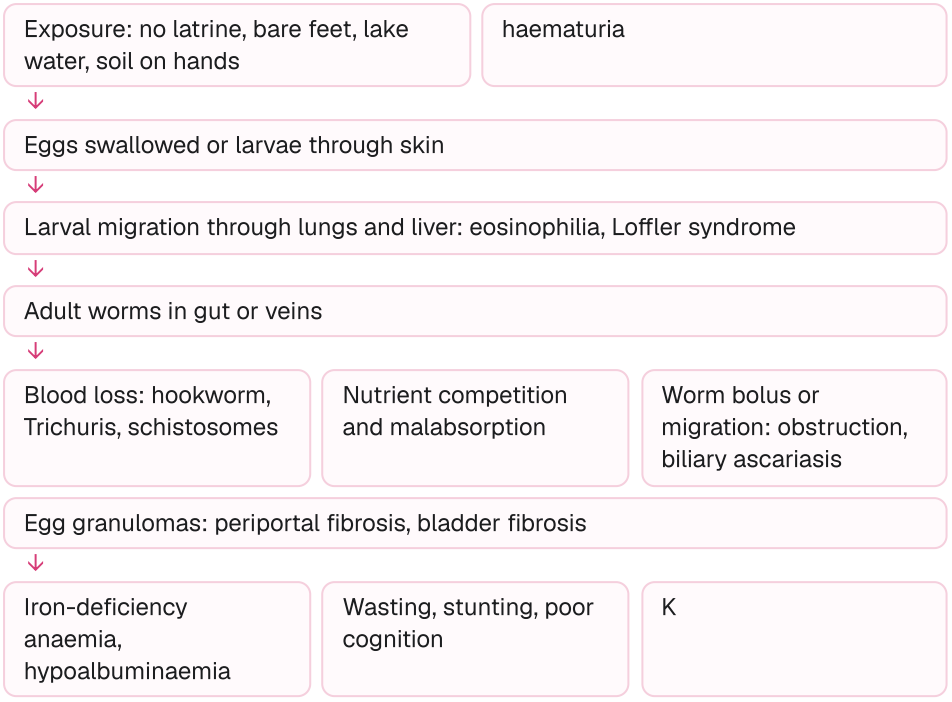
- **Ascaris**: egg swallowed → larva hatches in the small gut → through the gut wall → portal vein → liver → lungs (Löffler: cough, wheeze, eosinophils) → coughed up and swallowed → adult in the small gut.
- **Hookworm**: larva in soil → through bare skin (ground itch) → blood → lungs → swallowed → attaches to small-gut mucosa and sucks blood → eggs in stool → larvae in warm moist soil.
- **Strongyloides**: as hookworm, **plus** larvae hatch in the gut and re-enter through the gut wall or perianal skin (autoinfection) → infection lasts for decades; steroids or HTLV-1 → hyperinfection.
- **Schistosoma**: cercaria through skin in lake water → lungs → liver → adults pair in veins (*mansoni*: mesenteric; *haematobium*: bladder) → eggs lodge in the liver or bladder wall → granulomas → fibrosis.

Mechanisms of disease

Worm	Mechanism	Clinical consequence
Hookworm	Each *Necator* worm takes about **0.03 mL** of blood a day (*Ancylostoma* about 0.15–0.2 mL); chronic gut blood and protein loss	**Iron-deficiency anaemia**, hypoalbuminaemia (oedema), poor growth, poor cognition
Ascaris	Larval migration through lungs; adults compete for nutrients and impair absorption (fat, vitamin A, lactose); worms tangle into a bolus or migrate when irritated (fever, drugs, anaesthesia)	Löffler syndrome; malnutrition; **intestinal obstruction, volvulus, intussusception**; **biliary ascariasis** (colic, cholangitis, pancreatitis, liver abscess); appendicitis
Trichuris	Anterior end threaded into the colonic mucosa → inflammation and bleeding	Dysentery-like **Trichuris dysentery syndrome**, **rectal prolapse**, anaemia, clubbing, stunting
Strongyloides	Autoinfection; with low cell-mediated immunity (**steroids**, HTLV-1, malnutrition, transplant) larvae multiply massively and carry gut bacteria through the bowel wall	Larva currens (fast-moving itchy rash), chronic diarrhoea; **hyperinfection/disseminated disease with Gram-negative sepsis and meningitis** (often fatal)
Enterobius	Females lay eggs on perianal skin at night	**Nocturnal perianal itch**, poor sleep, vulvovaginitis, rarely appendicitis
Taenia	Adult tapeworm in the gut	Usually mild: passing segments, epigastric discomfort
***T. solium* eggs**	Larvae encyst in brain/muscle/eye; dying cysts provoke inflammation	**Neurocysticercosis**: seizures, headache, raised ICP
S. mansoni	Eggs trapped in the liver's portal tracts → granulomas → **periportal (Symmers' "pipe-stem") fibrosis** → **presinusoidal portal hypertension** with **preserved liver function**	Hepatosplenomegaly, varices, **haematemesis**, anaemia; in adolescents, "hepatosplenic schistosomiasis"

Worm	Mechanism	Clinical consequence
S. haematobium	Eggs in the bladder wall → granulomas, fibrosis, calcification	**Terminal haematuria**, dysuria, hydronephrosis, bladder cancer (adults), female genital schistosomiasis

Why eosinophilia? Eosinophils fight helminths in the **tissues**. They rise during **larval migration** (Ascaris and hookworm in the lungs, Strongyloides, schistosomes, *Toxocara*) and are often **normal when adult worms simply sit in the gut lumen**. Unexplained eosinophilia in a Ugandan child = worms until proven otherwise.



Feature links

- Larvae through bare skin → ground itch on the feet; *Strongyloides* autoinfection → larva currens.
- Larvae through the lungs → Löffler syndrome: cough, wheeze, fleeting infiltrates, eosinophilia.
- Hookworm blood sucking → chronic gut blood loss → iron deficiency → pallor, fatigue, pica, poor school performance, flow murmur, heart failure if severe.
- Hookworm protein loss → hypoalbuminaemia → oedema.
- Ascaris nutrient competition → malnutrition, vitamin A deficiency, stunting.
- Ascaris bolus → colicky pain, bilious vomiting, distension, mobile "sausage" mass, absolute constipation.
- Ascaris migration into the bile or pancreatic duct → RUQ pain, jaundice, cholangitis, pancreatitis (epigastric pain to the back).
- *Trichuris* in the colonic mucosa → bloody mucoid diarrhoea, tenesmus → rectal prolapse; chronic disease → clubbing.
- *Strongyloides* hyperinfection → larvae carry gut bacteria → Gram-negative sepsis and meningitis.
- Pinworm eggs laid on perianal skin at night → nocturnal itch, poor sleep, excoriation, vulvovaginitis.
- *T. solium* cysts in the brain → seizures, headache, raised ICP.
- *S. mansoni* eggs in portal tracts → periportal fibrosis → presinusoidal portal hypertension → splenomegaly, varices, haematemesis, with normal liver function.
- *S. haematobium* eggs in the bladder wall → terminal haematuria, dysuria → fibrosis → hydronephrosis.

5. Classification / Staging

By group and route: STH, pinworm, tapeworms and trematodes (table in section 1); skin-penetrators versus swallowed eggs (section 3).

Intensity of infection (WHO, by eggs per gram of stool, Kato-Katz)

Worm	Light	Moderate	Heavy
Ascaris	1–4,999	5,000–49,999	**≥50,000**
Trichuris	1–999	1,000–9,999	**≥10,000**
Hookworm	1–1,999	2,000–3,999	**≥4,000**
S. mansoni	1–99	100–399	**≥400**
S. haematobium	1–49 eggs/10 mL urine		**≥50 eggs/10 mL urine**

Morbidity rises with the number of worms, which is why mass treatment that lowers worm loads works even though reinfection occurs.

Community endemicity (sets the deworming frequency): STH prevalence ≥20% → yearly; ≥50% → twice yearly (WHO 2017). Schistosomiasis prevalence ≥10% → annual praziquantel from 2 years (WHO 2022).

6. Clinical Features

History

Question	Why it matters
Passed a worm (in stool, vomit, from the nose)? What did it look like?	Long pink worm = *Ascaris*; white flat segments = *Taenia*; tiny white threads around the anus = pinworm
Abdominal pain: colicky? vomiting? constipation? distension?	**Obstruction** from an *Ascaris* bolus
Right upper quadrant pain, jaundice, vomiting a worm	**Biliary ascariasis**, pancreatitis
Diarrhoea: bloody/mucoid, chronic? Rectal prolapse?	*Trichuris*, schistosomes, Strongyloides (also dysentery, IBD)
Night-time anal itch, poor sleep, vulval itch	**Pinworm**
Fatigue, pica (eating soil), breathlessness, poor school performance	**Anaemia** (hookworm); geophagia also exposes to eggs
Cough, wheeze 1–2 weeks after an itchy rash on the feet	Larval migration (Löffler, hookworm "ground itch")
Swimming/fishing/washing in the lake or river; where is home? school?	**Schistosomiasis** exposure
Blood at the end of urination	**S. haematobium**
Vomiting blood, big spleen	**S. mansoni portal hypertension**
Pork eating, pigs at home, seizures	**Cysticercosis**
Latrine at home? Shoes? Handwashing? Water source?	Exposure and reinfection risk

Question	Why it matters
Last deworming (date, drug), vitamin A, Child Days, school	Need for treatment now
Steroids planned or current; HIV status; malnutrition	**Strongyloides hyperinfection** risk
Growth and feeding; diet; immunisation card	Worms worsen malnutrition and iron deficiency; integrate care

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted for children: observe first, examine the abdomen while the child is calm on the parent's lap, and leave the perianal inspection (distressing) to the end.

Introduction, consent and preparation

- Introduce yourself; consent from the parent and assent from the child; **hand hygiene**; a chaperone for the perianal inspection.
- **Position and exposure**: supine for the abdomen with the child relaxed (knees bent); expose from nipples to knees, preserving dignity.

General inspection (end of the bed)

- Well or unwell; distress (colic, vomiting); **pallor**; jaundice; wasting or **oedema** (hypoalbuminaemia from hookworm, or kwashiorkor).
- **Vital signs**: HR (tachycardia, flow murmur in anaemia), RR, temperature, CRT; hydration if vomiting.
- **Growth**: weight, height/length, **Z-scores**, **MUAC**; plot on the chart (stunting).

Hands, face and neck

Sign	What it means
Palmar pallor, koilonychia	Iron-deficiency anaemia (hookworm, *Trichuris*, schistosomes)
Clubbing	Heavy *Trichuris*
Conjunctival pallor; jaundice	Anaemia; biliary ascariasis, cholangitis
Pale tongue	Anaemia
Raised JVP	Heart failure from severe anaemia
Subcutaneous nodules	Cysticercosis

Abdomen (the system)

- **Inspection**: distension, **visible peristalsis** (obstruction).
- **Palpation**: tenderness and guarding (peritonitis, volvulus), **mobile "rope-like" or "sausage" mass** (worm bolus), RUQ tenderness (biliary), **hepatomegaly (left lobe) and splenomegaly** (*S. mansoni*: measure the spleen in cm below the costal margin).
- **Percussion**: shifting dullness (ascites in portal hypertension); tympanic distension (obstruction).
- **Auscultation**: bowel sounds (active or tinkling in obstruction; absent in peritonitis).

Completing the examination

- **Chest**: wheeze, crackles (Löffler); flow murmur, gallop (severe anaemia).
- **Skin**: ground itch on the feet, **larva currens** (fast-moving serpiginous rash around buttocks/trunk: Strongyloides), cutaneous larva migrans (slow-moving, dog/cat hookworms).
- **Neurology**: seizures, focal signs, papilloedema (neurocysticercosis).
- **Perianal, last**: excoriation (pinworm), **rectal prolapse** (*Trichuris*); look for worms.

- **Bedside tests**: HemoCue, urine dipstick (section 8).
- **Thank** the parent and child, re-dress, **wash hands**, and **summarise**.
- **Document negatives**: "no abdominal mass, no organomegaly, no rectal prolapse, no pallor, MUAC 14.5 cm".

Red flags

RED FLAG

- **Colicky pain, bilious vomiting, distension, absolute constipation**: intestinal obstruction (worm bolus, volvulus, intussusception)
- **Peritonism, fever, shock**: perforation, gangrene, appendicitis
- **Jaundice, RUQ pain, fever**: biliary ascariasis with cholangitis; **epigastric pain radiating to the back**: pancreatitis
- **Haematemesis or melaena** with a big spleen: variceal bleeding from *S. mansoni*
- **Severe pallor** (Hb <7 g/dL), heart failure, oedema
- **A child on steroids (or about to start)** with diarrhoea, abdominal pain, cough or Gram-negative sepsis/meningitis: **Strongyloides hyperinfection**
- New seizures in a pork-eating community: neurocysticercosis (image before antiparasitics)

Thinking like a clinician

- **"Anaemic school child from a rural area"** → think **hookworm** (with malaria, poor diet and, at the lake, *S. mansoni*). Deworm **and** give iron, and check the response.
- **"Toddler vomiting worms with colicky pain and a mobile abdominal mass"** → **Ascaris bolus obstruction**. Do not give a standard anthelmintic dose as the first step; decompress and resuscitate first (section 9).
- **"Haematemesis with a huge spleen but normal liver function and no jaundice in a lakeside teenager"** → **presinusoidal portal hypertension from S. mansoni** (not cirrhosis).
- **"Terminal haematuria in a boy who swims in the river"** → **S. haematobium** (versus UTI, glomerulonephritis, stones). Check a midday urine.
- **"Steroids planned for nephrotic syndrome, asthma, TB meningitis or leukaemia"** → **treat empirically for Strongyloides** (ivermectin) first in endemic areas.
- **"Eosinophilia but stool microscopy negative"** → larval stage (too early for eggs), Strongyloides (needs special tests), schistosomes (eggs appear 5–8 weeks after exposure), *Toxocara*. Repeat stool, or treat empirically.
- **Common student mistakes**: treating anaemia with iron but not deworming (or the reverse); forgetting to repeat albendazole for pinworm and to treat the household; using single-dose albendazole for heavy *Trichuris*; giving ivermectin to a child under 15 kg; giving praziquantel/albendazole for suspected neurocysticercosis without imaging; forgetting Strongyloides before steroids.

EXAM FAVOURITE

Viva favourite: **"Why is Strongyloides different from the other STH?"** Answer: it can **complete its life cycle inside the host (autoinfection)**, so it persists for decades and can **multiply explosively** when immunity falls (**steroids, HTLV-1**), causing hyperinfection with Gram-negative sepsis and meningitis. It is **not detected by routine stool microscopy** (larvae, not eggs, and few of them) and **not treated by single-dose albendazole**; the drug of choice is **ivermectin**.

7. Differential Diagnoses

Differential (by presentation)	Distinguishing clinical feature	Distinguishing investigation
Iron-deficiency anaemia: hookworm, *Trichuris*, schistosomes vs poor diet, malaria, HIV, sickle cell disease, blood loss	Rural, barefoot, lakeside exposure; pica	FBC/film, stool, RDT, sickling test
Colicky pain and obstruction: *Ascaris* bolus vs **intussusception**, volvulus, incarcerated hernia, adhesions, typhoid perforation	Worms vomited or passed; mobile mass; redcurrant-jelly stool in intussusception	Ultrasound (whirlpool vs target sign), X-ray
Bloody diarrhoea / dysentery: heavy *Trichuris*, *S. mansoni* vs *Shigella*, amoebiasis, *Campylobacter*, IBD	Rectal prolapse, clubbing with *Trichuris*	Stool microscopy and culture
Rectal prolapse: *Trichuris* vs chronic constipation, cystic fibrosis, malnutrition, chronic diarrhoea	Bloody mucoid stool, anaemia	Stool microscopy
Haematuria: *S. haematobium* vs **UTI**, post-streptococcal GN, stones, trauma, sickle nephropathy, Wilms tumour	**Terminal** haematuria with water contact; no dysuria or oedema	Urine microscopy (eggs vs RBC casts), dipstick, USS
Hepatosplenomegaly: *S. mansoni* vs **hyperreactive malarial splenomegaly**, visceral leishmaniasis (Karamoja/north-east), SCD, leukaemia/lymphoma, chronic hepatitis B, storage disease	Lakeside home, left-lobe liver, normal liver function	USS (periportal fibrosis), FBC/film, malaria tests, LFTs
Perianal itch: *Enterobius* vs poor hygiene, candida, scabies, sexual abuse (in girls with vulvovaginitis, ask sensitively)	Nocturnal itch, household affected	Sticky-tape test
Cough, wheeze and eosinophilia: Löffler (Ascaris/hookworm larvae), Strongyloides vs asthma, TB, tropical pulmonary eosinophilia	Transient, after a foot rash; no atopy	FBC, CXR, stool (later)
Seizures: neurocysticercosis vs febrile seizures, epilepsy, meningitis, cerebral malaria, TB	Pork-eating community, pigs at home	CT/MRI

8. Clinical Workup

Key principles:

1. **You do not need a stool test to deworm.** In endemic areas, WHO and UCG treat **presumptively** (preventive chemotherapy) without individual diagnosis.
2. **Investigate the sick or atypical child**: anaemia not responding to treatment, chronic diarrhoea, haematuria, hepatosplenomegaly, eosinophilia, steroids planned, surgical signs.
3. **Know the blind spots of stool microscopy**: pinworm (use sticky tape), *Strongyloides* (larvae, few: use concentration/agar plate/serology), early infection (prepatent period), light infections (repeat 3 samples), *S. haematobium* (urine, not stool).

Bedside

Test	How	What it tells you	Available at
HemoCue Hb	Finger prick	Anaemia severity (hookworm, *Trichuris*, schisto)	HC III and above
Sticky-tape (Scotch tape) test	Press clear tape on the perianal skin **first thing in the morning before washing**, stick it on a slide; repeat on 3 mornings	Pinworm eggs (stool microscopy usually misses them)	HC III and above (slide read in the lab)
Urine dipstick	Midday urine	**Microscopic haematuria** strongly suggests *S. haematobium* in endemic areas	HC III and above

Test	How	What it tells you	Available at
Urine POC-CCA	Urine cassette (survey tool)	*S. mansoni* antigen	NTD programme surveys
Inspect any passed worm or segment	Bring it in a container	Species identification	All levels
Malaria RDT	Finger prick	Co-existing cause of anaemia/fever	All levels
MUAC, weight and height		Malnutrition linked to worms	All levels

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
Stool microscopy (direct wet mount)	Symptomatic child; diarrhoea; anaemia	Eggs of *Ascaris*, hookworm, *Trichuris*, *Taenia*, *S. mansoni* (lateral spine); *Strongyloides* larvae (if heavy)	Species-specific treatment	**HC III and above**
Urine microscopy (filtered, midday 10:00–14:00)	Haematuria, endemic area	*S. haematobium* eggs (terminal spine)	Praziquantel	HC III and above (filtration mainly in surveys)
Full blood count and film	Anaemia, systemic symptoms	**Microcytic hypochromic anaemia**; **eosinophilia** (>0.5 × 10^9/L)	Iron; supports tissue-migrating worms	HC IV and above
HIV test	Strongyloides, severe or recurrent infections	Positive	ART; Strongyloides risk	All levels

Imaging

Test	When	Expected finding	How it changes management	Available at
Abdominal ultrasound	Pain, mass, jaundice, hepatosplenomegaly, haematuria	**Worm bolus** ("bundle", "whirlpool" or "railway-track" sign), **worms in the bile duct**, **periportal fibrosis** (thick echogenic portal tracts), splenomegaly, bladder wall thickening/polyps, hydronephrosis	Surgery vs conservative; schisto staging	Most hospitals and many HC IVs
Plain abdominal X-ray (erect/supine)	Obstruction	Dilated loops, air–fluid levels, **whorled worm mass**	Obstruction management	General Hospital and above
CT/MRI brain	Seizures with suspected neurocysticercosis	Cysts (with scolex), ring-enhancing lesions, calcifications	Antiparasitic and steroid decisions; neurology referral	Kampala and some RRHs

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Formol-ether concentration, repeat stools (3 samples)	Light infections, eosinophilia, Strongyloides suspected	Increases sensitivity	Detects missed infections	Hospital labs
Kato-Katz	Surveys; quantifying intensity	Eggs per gram	Heavy infection = more morbidity	Mainly research/NTD programme
Agar-plate culture / Baermann technique	Suspected Strongyloides	Larvae	Ivermectin	Referral labs
Strongyloides serology	Before immunosuppression, eosinophilia	Positive	Treat	Limited (private/referral labs)
Serum albumin, LFTs	Oedema, hepatosplenomegaly	Low albumin (hookworm); normal liver enzymes and synthetic function in *S. mansoni* fibrosis	Distinguishes schisto from cirrhosis	RRH/NRH
Upper GI endoscopy	Haematemesis	Oesophageal varices	Band ligation/sclero therapy	Mulago, Lacor, some private and mission hospitals
ERCP	Persistent biliary ascariasis	Worm in the duct	Extraction	Mulago and private centres

Putting it together

Model one-line summary

> "A 9-year-old boy from Mayuge (Lake Victoria shore), weight 22 kg (weight-for-age Z −2.1, height-for-age Z −2.3: stunted), last dewormed 14 months ago, who swims and fishes daily, presenting with fatigue and pallor: Hb 6.8 g/dL (microcytic), hookworm and S. mansoni eggs on stool microscopy, spleen 4 cm, liver left lobe enlarged: polyparasitism with hookworm iron-deficiency anaemia and intestinal schistosomiasis with early hepatosplenic disease."

Problem list

1. Moderate–severe iron-deficiency anaemia.
2. Hookworm.
3. *S. mansoni* with hepatosplenomegaly (risk of portal hypertension).
4. Stunting.
5. Ongoing exposure (water contact, no shoes).

Model case presentation (ward round)

> "Master O, a 3-year-old boy from Wakiso, **weight 11.5 kg (weight-for-age Z −1.6), height 89 cm (height-for-age Z −1.3), MUAC 13.8 cm**, admitted 12 hours ago with 2 days of colicky abdominal pain, bilious vomiting and no stool, after vomiting a long pink worm at home. He has never been dewormed; the family uses a shared pit latrine and he plays barefoot. **Fully immunised for age** (card seen; MR2 given), last vitamin A 7 months ago; eating family foods; HIV status of mother negative (ANC). On admission: HR 140, RR 32, temperature 37.4 °C, CRT 2 s, mild dehydration, distended abdomen with a **mobile, non-tender**

> **sausage-shaped mass** in the periumbilical region, active bowel sounds, no guarding. Hb 9.8 g/dL, K^+ 3.1 mmol/L. **Ultrasound: a bolus of worms in the jejunum ('whirlpool' appearance), no free fluid; X-ray: dilated loops with air–fluid levels. Assessment: partial small-bowel obstruction from an Ascaris bolus, with mild dehydration and hypokalaemia. Management so far:** NBM, NGT on free drainage (two worms came up), IV fluids with potassium, analgesia, surgical review (conservative for now). **Plan:** reassess abdomen 4-hourly for peritonism; anthelmintic per the surgeon once obstruction settles; deworm the household; WASH counselling; growth follow-up."

EXAM FAVOURITE

Worked example 1: deworming doses by age and weight.

- 15-month-old (9 kg): **albendazole 200 mg** single dose (or mebendazole 250 mg).
- 4-year-old (15 kg): **albendazole 400 mg** single dose (or mebendazole 500 mg). Praziquantel for *S. mansoni*: **40 mg/kg = 600 mg** (one 600 mg tablet). Ivermectin is avoided under 15 kg in most guidance; at exactly 15 kg give it only if needed and with senior advice.
- 10-year-old (28 kg) with *Strongyloides*: **ivermectin 200 µg/kg = 5.6 mg** (≈ two 3 mg tablets = 6 mg) once daily for 1–2 days (CDC); UCG 2023 gives a single dose of 150 µg/kg (4.2 mg).

EXAM FAVOURITE

Worked example 2: reading a FBC. Hb 6.8 g/dL, MCV 58 fL, eosinophils 1.8 × 10^9/L (18%), platelets 480 × 10^9/L. **Interpretation:** microcytic anaemia (iron deficiency, with reactive thrombocytosis) **plus eosinophilia** (tissue-migrating helminths: hookworm larvae, schistosomes, Strongyloides). This fits hookworm and schistosomiasis; treat both, give iron, and recheck Hb in 4 weeks (expect a rise of about 1–2 g/dL per month with adequate iron).

9. Management

Immediate / emergency

Most children with worms are seen in OPD and are stable. Use this for the sick ones:

1. **ETAT / ABCDE**: airway (vomiting child, aspiration of worms), breathing, **shock** (dehydration from vomiting, sepsis, bleeding varices), coma, convulsions.
2. **Surgical abdomen?** Distension, bilious vomiting, peritonism → **nil by mouth, NGT on free drainage, IV access, bloods, IV fluids, surgical review now**.
3. **Bleeding?** Haematemesis → two IV lines, **group and cross-match**, Hb, resuscitate, refer for endoscopy (portal hypertension below).
4. **Severe anaemia?** HemoCue; Hb <4 g/dL or decompensated → transfusion.
5. **Seizure?** Diazepam 0.5 mg/kg PR (max 10 mg), glucose check.

Definitive

Routine deworming (preventive chemotherapy)

Group	Drug and dose	Frequency
Children 12–23 months	**Albendazole 200 mg** single dose, or **mebendazole 250 mg** (UCG)	Every 6 months (UCG 2023)

Group	Drug and dose	Frequency
Children ≥2 years	**Albendazole 400 mg** single dose, or **mebendazole 500 mg** single dose	Every 6 months (UCG); WHO: yearly if STH prevalence ≥20%, twice yearly if ≥50%
Under 12 months	Not routinely dewormed	Treat only a confirmed, symptomatic infection with senior advice
Adolescent girls, women of reproductive age, pregnant women (2nd/3rd trimester)	Albendazole 400 mg or mebendazole 500 mg	ANC (2nd trimester) and programmes
Schistosomiasis-endemic communities (prevalence ≥10%)	**Praziquantel 40 mg/kg** single dose, from **2 years** (WHO 2022)	**Annually** (WHO 2022); school-based in Uganda

How to give it: **chewable tablets** can be crushed and mixed with water for young children; never force a whole tablet into a small child's mouth (choking risk); **praziquantel with food** (reduces nausea), and in MDA the dose is estimated with a **height-based "dose pole"**.

> NOTE
> **The "worm wars".** A Cochrane review (Taylor-Robinson et al., 2019) found that **mass** deworming of all children in endemic areas produced **little or no average improvement** in weight, Hb, cognition or school attendance, while **treating children known to be infected** does improve weight and Hb. WHO and Uganda continue mass deworming because it is cheap and safe, heavily infected children (who benefit most) are hard to identify, and it reduces severe morbidity. For your viva: deworming is not a substitute for WASH, nutrition and iron.

Treatment of confirmed or symptomatic infections

- **Ascaris, hookworm**: single-dose albendazole (or mebendazole).
- **Trichuris** (symptomatic or heavy): albendazole for **3 days** (single dose cures poorly).
- **Pinworm**: single dose, **repeat after 2 weeks**, treat the **whole household**; cut nails, wash hands, change and wash bedding and underwear.
- **Strongyloides**: **ivermectin** (albendazole is the alternative); hyperinfection = stop or reduce steroids, treat Gram-negative sepsis/meningitis (ceftriaxone), ICU, refer.
- **Taenia, H. nana, schistosomiasis**: praziquantel at the species-specific dose.
- **Katayama fever** (acute schistosomiasis, non-immune child): praziquantel + prednisolone; repeat praziquantel after 6–8 weeks (immature worms are resistant). Rare in endemic children; seen in visitors.
- **Neurocysticercosis**: **specialist only**, after CT/MRI (UCG: refer); treating without imaging can provoke seizures and raised ICP.

Doses are in the drug table.

Drug table

Drug (indication)	Dose (paediatric; max)	Route	Frequency	Duration	Key cautions
Albendazole (routine; *Ascaris*, hookworm)	**200 mg** (12–23 months); **400 mg** (≥2 years)	Oral, chewed or crushed	Once	Single dose; routine every 6 months	Not routinely under 12 months; avoid in 1st trimester; worms may migrate out of mouth or nose (warn parents)

Drug (indication)	Dose (paediatric; max)	Route	Frequency	Duration	Key cautions
Mebendazole (routine; *Ascaris*, hookworm)	**250 mg** (1–2 years); **500 mg** (>2 years); or **100 mg** 12-hourly	Oral	Once, or 12-hourly	Single dose, or 3 days	As albendazole
Albendazole (*Trichuris*, symptomatic/heavy)	**400 mg**	Oral	**Once daily**	**3 days** (CDC)	Alternatively mebendazole 100 mg 12-hourly × 3 days; ivermectin 200 µg/kg × 3 days added in resistant cases
Albendazole (*Enterobius*)	**400 mg** (200 mg under 2)	Oral	Once	**Repeat after 2 weeks**	Treat the whole household
Ivermectin (*Strongyloides*, first line)	**200 µg/kg** (CDC); UCG 2023: 150–200 µg/kg	Oral, empty stomach	Once daily	**1–2 days** (CDC; UCG single dose)	Avoid **under 15 kg**, in pregnancy, and where *Loa loa* is endemic (not Uganda)
Albendazole (*Strongyloides*, alternative)	**400 mg**	Oral	**12-hourly**	**3 days** (UCG; CDC 7 days)	Use if < 15 kg
Ivermectin (*Strongyloides* hyperinfection)	**200 µg/kg**	Oral	Daily	Until stool/sputum negative for 2 weeks	Stop or reduce steroids; ceftriaxone for sepsis/meningitis; ICU; refer
Praziquantel (schistosomiasis, *mansoni* and *haematobium*)	**40 mg/kg** (UCG 2023; WHO)	Oral, with food	Once (can split into 2 doses 4–6 h apart)	Single dose; repeat in 4–6 weeks if symptoms persist	From **2 years**; abdominal pain, nausea, dizziness, urticaria (dying worms); arpraziquantel (dispersible) for 3 months–6 years being introduced
Praziquantel (*Taenia*)	**5–10 mg/kg**	Oral	Once	Single dose	Never for suspected cysticercosis without imaging
Praziquantel (*Hymenolepis nana*)	**25 mg/kg**	Oral	Once	Repeat after 10 days if needed	Autoinfection
Niclosamide (*Taenia*, UCG alternative)	**<2 years 500 mg; 2–6 years 1 g; >6 years 2 g**	Oral, chewed	Once	Single dose	Then bisacodyl 2 h later
Prednisolone (Katayama fever)	**1 mg/kg/day**	Oral	Daily	5 days	With praziquantel; repeat praziquantel at 6–8 weeks
Albendazole (neurocysticercosis, specialist only)	**15 mg/kg/day** (± praziquantel)	Oral	Daily	Specialist	With steroids and antiepileptics, **after CT/MRI**
Piperazine (Ascaris obstruction, where available)	About **75 mg/kg** (max 3.5 g)	Oral/NG	Once daily	2 days	Paralyses worms; per surgeon's protocol; then albendazole later
Elemental iron (+ folic acid)	**About 3 mg/kg/day** (WHO Pocket Book range 3–6 mg/kg/day; max about 200 mg/day)	Oral, between meals	Daily	**3 months**	Black stools; delay until acute malaria or severe infection treated

Drug (indication)	Dose (paediatric; max)	Route	Frequency	Duration	Key cautions
Whole blood / packed cells (Hb <4 g/dL or decompensated)	**20 mL/kg** whole blood (10 mL/kg packed cells)	IV	Over 3–4 h	Once	Slower and less in heart failure, with furosemide 1 mg/kg
Diazepam (seizure)	**0.5 mg/kg** (max 10 mg)	Rectal	Once; may repeat per seizure protocol		Check glucose

DRUG DOSING

Safety notes. Albendazole and mebendazole: mild abdominal pain, rarely worm migration through the mouth or nose (frightening for parents; warn them); avoid in the **first trimester** of pregnancy. **Ivermectin**: avoid **under 15 kg**, in pregnancy, and where *Loa loa* is endemic (not Uganda). **Praziquantel**: now recommended from **2 years**; a paediatric formulation (**arpraziquantel**, dispersible) for children **3 months to 6 years** has been developed and is being introduced in African programmes; check whether your district stocks it.

UGANDA CONTEXT

UCG 2023 vs CDC/WHO. UCG gives ivermectin **150 µg/kg as a single dose** (range 150–200) and albendazole for *Strongyloides* for 3 days; CDC uses **200 µg/kg for 1–2 days** and albendazole for 7 days. UCG deworms **every 6 months** regardless of prevalence; WHO 2017 sets the frequency by community prevalence (yearly ≥20%, twice yearly ≥50%). Quote UCG in Ugandan exams and name the international dose when asked.

Referral criteria

Level	Action	Refer up when
VHT / HC II	Routine deworming (albendazole/mebendazole), Child Days; counselling on WASH	Anaemia, abdominal pain with vomiting, haematuria, haematemesis → HC III/IV
HC III	Stool and urine microscopy, treatment of STH and pinworm, **Strongyloides (ivermectin)**, iron; praziquantel for tapeworm (UCG level HC3)	**Surgical abdomen** (refer urgently after NGT, IV fluids), severe anaemia needing transfusion, schistosomiasis with hepatosplenomegaly → HC IV / General Hospital
HC IV / General Hospital	**Praziquantel** (UCG level HC4 for schistosomiasis), USS, transfusion, surgery for obstruction in some HC IVs	Obstruction needing paediatric surgery, variceal bleeding, biliary ascariasis not settling, seizures with suspected neurocysticercosis, Strongyloides hyperinfection → RRH
RRH / NRH	Paediatric surgery, ERCP, endoscopy for varices (Mulago, Lacor), CT/MRI for neurocysticercosis, ICU, specialist care	n/a

Monitoring and follow-up

- **Anaemia**: recheck Hb at **4 weeks** (expect a rise of about 1–2 g/dL); no rise → poor adherence, ongoing blood loss, malaria, HIV, sickle cell disease, thalassaemia trait.
- **Obstruction**: reassess the abdomen **4-hourly** for peritonism; surgery if not resolving within 24–48 h.
- **Pinworm**: second dose at 2 weeks; household treated.
- **Schistosomiasis**: repeat praziquantel in 4–6 weeks if symptoms persist; follow spleen size and ultrasound in hepatosplenic disease.
- **Routine**: next deworming in 6 months; growth monitoring; vitamin A.

Iron-deficiency anaemia from worms

1. **Deworm** (and treat schistosomiasis if present).
2. **Oral iron**: elemental iron **about 3 mg/kg/day** (WHO Pocket Book range 3–6 mg/kg/day; max about 200 mg/day) **for 3 months** (to refill stores), with **folic acid**; give between meals with fruit (vitamin C); warn about black stools. Delay iron until **after acute malaria or severe infection** is treated.
3. **Diet**: beans, green leafy vegetables, liver, small fish (mukene), eggs.
4. **Transfuse** only for **Hb <4 g/dL** or decompensation (heart failure, respiratory distress): whole blood 20 mL/kg (10 mL/kg packed cells) over 3–4 h; slowly (and less) if heart failure, with furosemide 1 mg/kg.
5. **Recheck Hb at 4 weeks**: expect a rise of about 1–2 g/dL.

Ascaris intestinal obstruction (algorithm)

1. **Nil by mouth**, **NGT** on free drainage (worms may come up), **IV access**.
2. **Resuscitate**: correct dehydration and electrolytes (hypokalaemia), maintenance fluids.
3. **Bloods**: Hb, electrolytes, group and save; **abdominal USS and/or erect X-ray**.
4. **Surgical review early**.
5. **Partial obstruction, no peritonism → conservative** for 24–48 h: NGT, IV fluids, analgesia, hypertonic saline or gastrografin enema in some units. Many paediatric surgeons give the anthelmintic **only once the obstruction is settling**: classically **piperazine** (paralyses worms so they pass passively; about 75 mg/kg once daily for 2 days, max 3.5 g, where available), then albendazole later. Giving albendazole or mebendazole during complete obstruction may stimulate the worms and worsen the bolus. **Follow your surgeon's protocol.**
6. **Surgery (laparotomy)** if: **peritonitis, perforation, volvulus, intussusception, complete obstruction not resolving within 24–48 h, or gangrene**. Options: milking the worms into the colon, enterotomy, resection.
7. After recovery: deworm the child and the household; treat malnutrition; address sanitation.

Biliary and pancreatic ascariasis

Conservative first: NBM, IV fluids, analgesia, antispasmodic, antibiotics if cholangitis (e.g. ceftriaxone + metronidazole), anthelmintic once able to take oral drugs (most worms leave the duct). **ERCP** extraction if symptoms persist or a dead worm remains in the duct (Mulago and private centres); surgery rarely.

Schistosomal portal hypertension

1. **Acute variceal bleed**: ABC, two IV lines, **cross-match**, transfuse to Hb about 7–8 g/dL (avoid over-transfusion, which raises portal pressure), **vitamin K**, antibiotics (e.g. ceftriaxone), octreotide if available, **urgent referral for endoscopic band ligation or sclerotherapy** (Mulago, Lacor).
2. **Praziquantel 40 mg/kg** once stable (early periportal fibrosis can regress in children).
3. **Prevent rebleeding**: propranolol (specialist), repeat endoscopy; nutrition, iron.
4. Liver function is usually preserved (unlike cirrhosis), so outcomes after controlling bleeding are reasonable.

10. Complications

Disease-related

Worm	Complications
All STH (heavy, chronic)	Iron-deficiency anaemia, **stunting**, wasting, poor cognition and school performance, vitamin A deficiency

Worm	Complications
Ascaris	**Intestinal obstruction**, volvulus, intussusception, perforation, **biliary ascariasis**, cholangitis, **pancreatitis**, liver abscess, appendicitis, aspiration of a migrating worm, Löffler pneumonitis
Hookworm	Severe anaemia, heart failure, hypoalbuminaemic oedema, growth and cognitive delay
Trichuris	Trichuris dysentery syndrome, **rectal prolapse**, anaemia, clubbing, stunting
Strongyloides	Chronic malabsorption; **hyperinfection/dissemination** with Gram-negative sepsis and meningitis (mortality often over 50%)
Enterobius	Excoriation and secondary bacterial infection, vulvovaginitis, UTI in girls, rarely appendicitis
Taenia solium	**Neurocysticercosis** (epilepsy, hydrocephalus, raised ICP), ocular and muscular cysticercosis
S. mansoni	Periportal fibrosis, **portal hypertension**, **variceal bleeding**, splenomegaly/hypersplenism, ascites, anaemia, stunting
S. haematobium	Haematuria, anaemia, bladder fibrosis/calcification, **hydronephrosis**, renal failure, female genital schistosomiasis (later HIV risk, infertility), **squamous cell bladder cancer** in adults

Treatment-related

Complication	Cause	Recognition and action
Worse obstruction	Albendazole/mebendazole during complete Ascaris obstruction	Decompress and resuscitate first; anthelmintic per surgeon
Worm migration out of mouth or nose; aspiration	Anthelmintic irritating worms	Warn parents; protect the airway in a vomiting child
Seizures, raised ICP	Antiparasitic treatment of neurocysticercosis without imaging and steroid cover	Image first; specialist only
Strongyloides hyperinfection	Steroids started without treating Strongyloides	Ivermectin before steroids in endemic areas
Abdominal pain, nausea, dizziness, urticaria	Praziquantel (dying worms)	Give with food; reassure
Choking	Whole tablet forced into a small child	Crush or chew
Raised portal pressure, rebleeding	Over-transfusion in variceal bleeding	Transfuse to Hb about 7–8 g/dL

ll. Prognosis

- Most light infections are cured by a single dose, but **reinfection within months** is the rule without WASH; morbidity tracks worm load, so regular treatment keeps it low.
- Anaemia from hookworm responds to deworming plus 3 months of iron (Hb rise about 1–2 g/dL per month).
- **Early periportal fibrosis from *S. mansoni* can regress in children** after praziquantel; established portal hypertension has reasonable outcomes once bleeding is controlled, because liver function is preserved.
- **Strongyloides hyperinfection** carries a mortality often over 50%.
- Chronic heavy infection in early childhood contributes to stunting and poorer cognition; how much mass deworming alone reverses this is debated (Cochrane 2019).

12. Prevention

Measure	Why
Latrines used by everyone, including children (no open defecation)	Breaks every faecal–oral and soil cycle
Handwashing with soap after the toilet and before eating; cut nails	*Ascaris*, *Trichuris*, pinworm, *T. solium* eggs
Wear shoes	Hookworm, Strongyloides larvae enter through bare feet
Wash and cook vegetables; cook pork and beef thoroughly; confine pigs	*Ascaris*; tapeworms; cysticercosis
Avoid swimming, bathing and washing in lakes and slow rivers; use borehole or tap water; don't urinate or defecate in water	Schistosomiasis (UCG 2023)
Regular deworming (Child Days, schools, IMCI, ANC)	Keeps worm loads low
Health education in schools	Children drive transmission and can change habits

13. Special Populations

- **Under 12 months**: not routinely dewormed; treat only a confirmed, symptomatic infection with senior advice.
- **Under 15 kg**: avoid ivermectin; use albendazole 400 mg 12-hourly for 3 days for *Strongyloides*.
- **Under 2 years**: praziquantel is now given from 2 years; arpraziquantel (3 months–6 years) where stocked.
- **Children about to start steroids** (nephrotic syndrome, asthma, TB meningitis, leukaemia): empirical ivermectin (or albendazole if < 15 kg) in endemic areas.
- **HIV, HTLV-1, malnutrition**: higher Strongyloides hyperinfection risk.
- **Adolescent girls and pregnant women**: deworm in the 2nd or 3rd trimester only; not in the first trimester; no ivermectin in pregnancy.
- **Non-immune visitors to lakes**: Katayama fever (praziquantel + prednisolone, repeat at 6–8 weeks).
- **Severe acute malnutrition**: deworm in the rehabilitation phase (see Malnutrition topic).

14. Key Points

High-yield points

- **Albendazole 200 mg (12–23 months) / 400 mg (≥2 years) every 6 months** from 1 year.
- **No stool test needed to deworm** in endemic areas.
- **Hookworm = iron-deficiency anaemia**: deworm **and** give iron for 3 months.
- **Ascaris obstruction**: NBM, NGT, IV fluids, surgeon first; anthelmintic later.
- **Trichuris**: 3 days of albendazole.
- **Pinworm**: sticky tape; repeat at 2 weeks; treat the household.
- **Strongyloides**: autoinfection, missed by microscopy, **ivermectin**, treat before steroids.
- **Praziquantel 40 mg/kg** for schistosomiasis; image before treating cysticercosis.
- **Lakeside child with haematemesis and a big spleen** = *S. mansoni* portal hypertension with normal liver function.
- **Eosinophilia = tissue migration.**

Numbers & doses

Item	Value
Routine deworming	**Every 6 months from 1 year** (UCG); WHO yearly (≥20%) / twice yearly (≥50% prevalence)
Albendazole	**200 mg** (12–23 months); **400 mg** (≥2 years) single dose
Mebendazole	**250 mg** (1–2 years); **500 mg** (>2 years) single dose; or 100 mg 12-hourly × 3 days
Trichuris (heavy)	Albendazole 400 mg daily × **3 days**
Pinworm	Albendazole single dose, **repeat at 2 weeks**, treat household
Ivermectin (*Strongyloides*)	**200 µg/kg** once daily × 1–2 days (CDC); UCG 150 µg/kg single dose; **not under 15 kg**
Albendazole for *Strongyloides* (alternative)	400 mg 12-hourly × 3 days (UCG)
Praziquantel	**40 mg/kg** schistosomes; **5–10 mg/kg** *Taenia*; **25 mg/kg** *H. nana*; from **2 years**
Niclosamide (*Taenia*)	<2 y **500 mg**; 2–6 y **1 g**; >6 y **2 g**
Iron	Elemental iron **~3 mg/kg/day × 3 months**
Transfusion threshold	Hb **<4 g/dL** or decompensated
Blood loss per hookworm	*Necator* ~0.03 mL/day; *Ancylostoma* ~0.15–0.2 mL/day
Heavy intensity (eggs/g)	*Ascaris* ≥50,000; *Trichuris* ≥10,000; hookworm ≥4,000; *S. mansoni* ≥400
S. haematobium urine	Collect **10:00–14:00** (midday); heavy ≥50 eggs/10 mL
Eosinophilia	>**0.5 × 10^9/L**
Global burden	~**1.5 billion** STH infections (24%); schisto PC needed by ~**254 million** (2024)

Exam traps

- **"Give albendazole to every child with Ascaris obstruction immediately."** Not as the first step: **decompress and resuscitate**; anthelmintic per the surgeon once the obstruction settles.
- **"A negative stool test excludes Strongyloides."** Wrong: routine microscopy misses it; use concentration, agar plate or serology, or treat empirically before steroids.
- **"Albendazole single dose clears whipworm."** Poor cure rates: give **3 days**.
- **"Ivermectin for a 10 kg toddler."** Avoid **under 15 kg**.
- **"Pinworm is diagnosed by stool microscopy."** Wrong: **sticky-tape test**; eggs are laid on the perianal skin.
- **"Haematemesis plus splenomegaly = cirrhosis."** In a lakeside child think **S. mansoni presinusoidal portal hypertension** with normal liver function.
- **"Treat neurocysticercosis with praziquantel at the HC IV."** Wrong: **image first**; treatment can trigger seizures and raised ICP (UCG: refer).
- **"Iron alone will correct this child's anaemia."** Not if hookworm keeps causing blood loss: **deworm too**.

Questions seniors ask

Viva questions

1. *Routine deworming doses in Uganda?* Albendazole 200 mg (12–23 months) or 400 mg (≥2 years), or mebendazole 250/500 mg, every 6 months.
2. *Which worm causes iron-deficiency anaemia most?* Hookworm (blood-sucking; *Ancylostoma* more than *Necator*).

3. *How do you diagnose pinworm?* Sticky-tape test in the morning before washing.
4. *Which worm can cause disease decades after leaving an endemic area, and why? Strongyloides*, through autoinfection.
5. *Praziquantel dose for schistosomiasis?* 40 mg/kg single dose.
6. *Why is the liver function normal in S. mansoni portal hypertension?* The fibrosis is periportal (presinusoidal); hepatocytes are spared.
7. *What is Löffler syndrome?* Transient eosinophilic pneumonitis (cough, wheeze, fleeting infiltrates) during larval migration of *Ascaris* (or hookworm) through the lungs.

On the ward round

- **This child needs prednisolone for nephrotic syndrome. What will you do first about worms?** In an endemic area, give **ivermectin 200 µg/kg** (if ≥15 kg; albendazole 400 mg 12-hourly × 3 days if smaller) to prevent Strongyloides hyperinfection, plus routine deworming.
- **Why do we deworm without testing?** Prevalence is high, drugs are cheap and safe, and individual testing is costly and insensitive; WHO recommends preventive chemotherapy in endemic areas.
- **How does hookworm cause anaemia, and how will you treat it?** Chronic intestinal blood loss (each worm sucks blood) → iron deficiency; treat with albendazole and **iron 3 mg/kg/day for 3 months**, and recheck Hb at 4 weeks.
- **What does a 'whirlpool' on ultrasound mean in a vomiting toddler?** A bolus of *Ascaris* worms (or volvulus): obstruction; NBM, NGT, IV fluids, surgical review.
- **How would you distinguish S. haematobium haematuria from a UTI?** Terminal haematuria in a child with water contact, with eggs (terminal spine) in a midday urine and no pyuria/nitrites; UTI has dysuria, pyuria and positive culture.
- **Why is reinfection so common after deworming?** Drugs don't prevent reinfection; eggs persist in soil for months; without latrines, handwashing, shoes and safe water, children are reinfected within months, which is why treatment is repeated every 6 months.

Mnemonics

- **Skin-penetrators: "SHS"**: **S**trongyloides, **H**ookworm, **S**chistosoma (shoes and avoiding lake water prevent them).
- **Swallowed eggs: "PATCH"**: **P**inworm, **A**scaris, **T**richuris, **C**ysticercosis (T. solium eggs), **H**ymenolepis.
- **Spines**: ***mansoni*** = **"M for Middle/lateral"** (lateral spine, mesenteric veins, in stool); ***haematobium*** = **"H for Head/terminal"** (terminal spine, bladder, in urine).
- **Albendazole "2-4"**: 200 mg under 2 years, 400 mg from 2 years.
- **Praziquantel "40 for flukes, 5–10 for tapes, 25 for dwarfs"**: 40 mg/kg schistosomes, 5–10 mg/kg *Taenia*, 25 mg/kg *H. nana*.
- **Pinworm "treat twice, treat the house"**: repeat at 2 weeks; all family members.

OSCE tips

- **Counselling station on deworming**: explain **why** (anaemia, growth, school), the **dose** (chew/crush for small children), **side effects** (mild pain; worms may come out), **every 6 months**, and **WASH** messages (latrine, handwashing, shoes, boiled water, no swimming in the lake).
- **Data-interpretation station (egg pictures)**: *Ascaris* (round, mammillated shell), *Trichuris* (barrel/lemon shape with two polar plugs), hookworm (thin shell, segmented embryo), *S. mansoni* (**lateral spine**), *S. haematobium* (**terminal spine**), *Taenia* (radially striated shell), *Enterobius* (flattened on one side, D-shaped).
- **Abdominal examination station**: in Hutchison order, measure the spleen and left lobe of the liver, look for a mobile mass, and finish with perianal inspection with a chaperone.

15. References

1. Ministry of Health Uganda. Uganda Clinical Guidelines 2023 (sections 2.4.1 Intestinal worms, 2.4.1.1 Taeniasis, 2.4.6 Schistosomiasis; 17.3.8 routine deworming). Kampala: MoH; 2023. https://library.health.go.ug/uganda-clinical-guidelines-2023
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SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Routine deworming in Uganda (UCG 2023)?	Every **6 months from 1 year**: **albendazole 200 mg (12–23 months) / 400 mg (≥2 years)** single dose, or **mebendazole 250 mg (1–2 years) / 500 mg (>2 years)**.
2	WHO thresholds for STH preventive chemotherapy?	**Once yearly** where prevalence is **≥20%**; **twice yearly** where **≥50%**; for preschool and school-age children, adolescent girls and women of reproductive age (WHO 2017).
3	Name the soil-transmitted helminths.	***Ascaris lumbricoides*** (roundworm), ***Trichuris trichiura*** (whipworm), **hookworms** (*Necator americanus*, *Ancylostoma duodenale*) and ***Strongyloides stercoralis***.
4	Which worms penetrate the skin?	**Hookworm, Strongyloides, Schistosoma** (mnemonic SHS) — prevented by shoes and avoiding infested water.
5	How does hookworm cause anaemia?	Adults attach to the small-intestinal mucosa and **suck blood** (*Necator* ~0.03 mL/day, *Ancylostoma* ~0.15–0.2 mL/day per worm) → chronic blood and protein loss → **iron-deficiency anaemia** and hypoalbuminaemia.
6	What is Löffler syndrome?	Transient **eosinophilic pneumonitis** (cough, wheeze, fleeting infiltrates, eosinophilia) as *Ascaris* (or hookworm) larvae migrate through the lungs.
7	Surgical complications of Ascaris?	**Intestinal obstruction** (worm bolus), volvulus, intussusception, perforation, **biliary ascariasis** (colic, cholangitis), **pancreatitis**, liver abscess, appendicitis.
8	Features of heavy Trichuris infection?	**Trichuris dysentery syndrome** (bloody mucoid diarrhoea), **rectal prolapse**, anaemia, **finger clubbing**, stunting. Treat with **albendazole 400 mg daily × 3 days**.
9	Why is Strongyloides special?	**Autoinfection** (persists for decades), **missed by routine stool microscopy**, not cleared by single-dose albendazole, and **steroids/HTLV-1** trigger **hyperinfection** with Gram-negative sepsis/meningitis. Drug of choice: **ivermectin**.
10	Ivermectin dose for Strongyloides?	**200 µg/kg** orally once daily for 1–2 days (CDC); UCG 2023 150 µg/kg single dose. **Avoid under 15 kg.** Alternative: albendazole 400 mg 12-hourly × 3 days (UCG).
11	What is larva currens?	A **fast-moving (cm per hour) serpiginous itchy rash** around the buttocks, perineum and trunk from migrating *Strongyloides* larvae (autoinfection).
12	Pinworm: symptoms, diagnosis and treatment?	**Night-time perianal itch**, poor sleep, vulvovaginitis. **Sticky-tape test** in the morning before washing. **Albendazole 400 mg (200 mg under 2) once, repeat after 2 weeks**; treat the whole household; hygiene.
13	Praziquantel doses?	**Schistosomiasis 40 mg/kg** single dose; **Taenia 5–10 mg/kg** single dose; ***Hymenolepis nana* 25 mg/kg** single dose. Give with food; from 2 years.
14	Niclosamide doses for tapeworm (UCG)?	**<2 years 500 mg; 2–6 years 1 g; >6 years 2 g**, chewed, single dose, followed by bisacodyl 2 hours later.

#	Question	Answer
15	What causes neurocysticercosis?	Swallowing ***Taenia solium* eggs** (faecal–oral from a human tapeworm carrier), not eating pork itself; larvae encyst in the brain → **seizures**, headache, hydrocephalus. **Image before treating** (UCG: refer).
16	S. mansoni vs S. haematobium?	***mansoni***: mesenteric veins, **lateral spine** eggs in stool, periportal fibrosis, portal hypertension. ***haematobium***: bladder venous plexus, **terminal spine** eggs in **midday urine**, terminal haematuria, hydronephrosis, bladder cancer.
17	Why is liver function preserved in schistosomal portal hypertension?	Eggs cause **periportal (Symmers' pipe-stem) fibrosis → presinusoidal** portal hypertension; hepatocytes are spared, so albumin and clotting are usually normal (unlike cirrhosis).
18	Where is S. mansoni endemic in Uganda?	Shores of **Lake Victoria, Lake Albert, Lake Kyoga** and the **Albert Nile** (fishing communities, e.g. Mayuge, Namayingo, Buliisa, Pakwach).
19	WHO 2022 schistosomiasis guideline: preventive chemotherapy?	**Annual praziquantel 40 mg/kg** for all age groups **from 2 years** (including adults, pregnant women after the first trimester and lactating women) in communities with prevalence **≥10%**.
20	When collect urine for S. haematobium eggs?	**Midday (10:00–14:00)**, when egg excretion peaks; filter 10 mL (heavy = ≥50 eggs/10 mL). Microscopic haematuria on dipstick is a good screening proxy.
21	What does eosinophilia indicate in a child with worms?	**Tissue-migrating** stages: Ascaris/hookworm larvae, Strongyloides, schistosomes, Toxocara; adults sitting in the gut lumen often give **normal** eosinophils.
22	Iron treatment for hookworm anaemia?	Deworm, then **elemental iron ~3 mg/kg/day for 3 months** with folic acid (after acute malaria/infection is treated); recheck Hb at 4 weeks (expect +1–2 g/dL).
23	Initial management of Ascaris bolus obstruction?	**NBM, NGT on free drainage, IV fluids and electrolytes, analgesia, early surgical review**; conservative for 24–48 h if no peritonism; anthelmintic per surgeon once settling; **laparotomy** for peritonitis, volvulus, intussusception or failure.
24	Age limits for deworming drugs?	**Albendazole/mebendazole**: not routinely under 12 months. **Ivermectin**: avoid under 15 kg. **Praziquantel**: from 2 years (paediatric arpraziquantel for 3 months–6 years being introduced).
25	Kato-Katz heavy-intensity thresholds (eggs per gram)?	*Ascaris* **≥50,000**; *Trichuris* **≥10,000**; hookworm **≥4,000**; *S. mansoni* **≥400**.
26	Key WASH measures against worms?	Latrine use, **handwashing with soap**, cutting nails, **wearing shoes**, washing/cooking food, cooking pork and beef well, safe water, **avoiding swimming/washing in lakes and slow rivers**.
27	Identify the eggs: lemon-shaped with two polar plugs; round with a bumpy shell; D-shaped flattened on one side.	**Trichuris** (barrel/lemon with plugs); **Ascaris** (round, mammillated shell); **Enterobius** (D-shaped, flattened on one side).

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. An 8-year-old girl (21 kg) from Luweero is brought for tiredness and poor school performance. She walks barefoot and the family has no latrine. She is pale, HR 110, no heart failure. Hb 7.2 g/dL, MCV 61 fL, eosinophils 1.2 x 10^9/L. Malaria RDT negative. She was last dewormed 2 years ago. What is the MOST appropriate management?

A Transfuse whole blood 20 mL/kg today
B Oral iron alone for 3 months
C Praziquantel 40 mg/kg only
D Albendazole 400 mg single dose plus oral elemental iron about 3 mg/kg/day for 3 months, WASH counselling, recheck Hb in 4 weeks
E Ivermectin 200 µg/kg for 2 days only

2. A 3-year-old boy (11.5 kg) vomited a long pink worm yesterday. He now has colicky abdominal pain, bilious vomiting and no stool for 2 days. HR 140, CRT 2 s, abdomen distended with a mobile periumbilical sausage-shaped mass, no guarding. Ultrasound shows a bolus of worms. What is the BEST initial management?

A Albendazole 400 mg orally now and discharge
B Nil by mouth, nasogastric tube on free drainage, IV fluids with potassium, and urgent surgical review
C Immediate laparotomy
D Oral mebendazole 100 mg 12-hourly for 3 days at home
E Barium meal to confirm the diagnosis

3. A 6-year-old boy (19 kg) with newly diagnosed nephrotic syndrome is about to start prednisolone 60 mg/m2/day at a regional referral hospital in eastern Uganda. He has had intermittent loose stools and an itchy rash on his buttocks. Eosinophils are 1.4 x 10^9/L; stool wet prep is negative. What should be done before starting steroids?

A Nothing; a negative stool test excludes worms
B Delay steroids for 3 months
C Praziquantel 40 mg/kg
D Repeat the stool wet prep weekly for a month
E Give ivermectin 200 µg/kg (about 4 mg) and then start prednisolone

4. A 5-year-old girl (17 kg) wakes every night scratching her bottom and has developed vulval itching. Her 2 siblings have similar symptoms. Which is the BEST diagnostic test?

A Sticky-tape test of the perianal skin first thing in the morning before washing
B Stool microscopy for ova
C Full blood count for eosinophilia
D Midday urine microscopy
E Abdominal ultrasound

5. For the same pinworm-infected 5-year-old (17 kg) and her family, which treatment plan is CORRECT?

A Praziquantel 40 mg/kg once for the child only

B Ivermectin for the child only

C Albendazole 400 mg for the child and family members (200 mg for any child aged 12–23 months), repeated after 2 weeks, plus nail-cutting, handwashing and washing bedding

D Albendazole 400 mg daily for 3 days for the child only

E Antihistamines alone

6. A 12-year-old boy (30 kg) from Buliisa on the shore of Lake Albert presents with vomiting about 200 mL of fresh blood. He fishes with his father daily. HR 128, BP 92/60, pale. Spleen 8 cm below the costal margin, liver left lobe palpable, no jaundice, no ascites. Albumin 38 g/L, INR 1.1. What is the likeliest cause of the bleeding?

A Peptic ulcer from NSAIDs

B Alcoholic cirrhosis

C Hyperreactive malarial splenomegaly with thrombocytopenia

D Oesophageal varices from presinusoidal portal hypertension due to Schistosoma mansoni

E Hepatitis B cirrhosis

7. For the boy in the previous question, after resuscitation and endoscopic band ligation at Mulago, which antiparasitic treatment is indicated?

A Albendazole 400 mg single dose

B Praziquantel 40 mg/kg (1.2 g) single dose once stable

C Ivermectin 200 µg/kg

D Niclosamide 2 g

E No antiparasitic treatment because fibrosis is irreversible

8. A 10-year-old boy (26 kg) from a district with S. haematobium foci reports blood in his urine at the end of urination for 3 months. He swims in a slow river after school. No dysuria, BP 104/66, no oedema. Dipstick: blood 3+, protein trace, leucocytes negative, nitrites negative. What is the BEST next test?

A Urine culture for E. coli

B Renal biopsy

C Stool Kato-Katz

D ASO titre and complement

E Microscopy of a filtered midday urine sample for eggs with a terminal spine

9. A 7-year-old girl (22 kg) has had bloody mucoid diarrhoea for 2 months, tenesmus, and her rectum prolapsed twice this week. She is pale (Hb 8.1 g/dL) and has finger clubbing. Stool shows numerous barrel-shaped eggs with polar plugs. What is the MOST appropriate treatment?

A Albendazole 400 mg once daily for 3 days, iron, and gentle reduction of the prolapse

B Albendazole 400 mg single dose

C Ciprofloxacin for shigellosis

D Metronidazole for amoebiasis

E Immediate rectopexy

10. A 9-year-old girl (25 kg) from a pig-rearing village has had three focal seizures with secondary generalisation over a month. She is well between seizures and has no fever. Her father passes 'white flat pieces' in his stool. What is the BEST next step?

A Praziquantel 40 mg/kg immediately at the health centre
B Albendazole 400 mg single dose and discharge
C Brain imaging (CT or MRI) and referral, with anticonvulsants; treat the father's tapeworm with praziquantel
D Lumbar puncture and ceftriaxone
E Reassure: febrile seizures

11. A mother brings her 9-month-old (8 kg) to a Child Days Plus outreach asking for 'worm medicine' and vitamin A. The baby is well, breastfeeding, and growing along the 0 Z-score line. What should the nurse give?

A Albendazole 200 mg and vitamin A 200,000 IU
B Albendazole 400 mg and vitamin A 100,000 IU
C Mebendazole 500 mg only
D Vitamin A 100,000 IU only; start deworming at 12 months
E Praziquantel 40 mg/kg and vitamin A 100,000 IU

12. A 4-year-old boy (14 kg) from Mayuge has hookworm and S. mansoni eggs on stool microscopy. He is well apart from mild pallor (Hb 9.6 g/dL). Which treatment combination is CORRECT?

A Ivermectin 2.8 mg plus albendazole 400 mg
B Albendazole 400 mg single dose plus praziquantel 40 mg/kg (560 mg) single dose with food, and iron
C Praziquantel 5 mg/kg plus mebendazole 100 mg
D Albendazole 200 mg plus niclosamide 1 g
E Mebendazole 500 mg alone because it treats all worms

13. A 2-year-old (12 kg) develops cough, wheeze and low-grade fever 10 days after his mother noticed an itchy rash on his feet. Chest X-ray shows patchy migratory infiltrates; eosinophils 2.1 x 10^9/L. Stool microscopy is negative. What is the likeliest explanation?

A Bacterial pneumonia
B Acute asthma
C Pulmonary tuberculosis
D Foreign-body aspiration
E Löffler syndrome from larval migration through the lungs (hookworm or Ascaris), before eggs appear in the stool

14. A 6-year-old girl (18 kg) presents with severe colicky right upper quadrant pain, vomiting and mild jaundice 3 days after receiving albendazole at school. Ultrasound shows a tubular structure with a central echogenic line in a dilated common bile duct. Temperature 37.8 °C. What is the likeliest diagnosis?

A Biliary ascariasis
B Acute viral hepatitis
C Gallstones from sickle cell disease
D Hydatid cyst rupture
E Acute appendicitis

15. Which statement about Strongyloides stercoralis is CORRECT?

A It is reliably diagnosed by a single routine stool wet prep
B Single-dose albendazole mass treatment eliminates it
C It can persist for decades through autoinfection and cause fatal hyperinfection when steroids are given
D It is acquired by eating undercooked pork
E It causes terminal haematuria

Answers & explanations

1. D · Albendazole 400 mg single dose plus oral elemental iron about 3 mg/kg/day for 3 months, WASH counselling, recheck Hb in 4 weeks

D is correct: microcytic anaemia with eosinophilia in a barefoot child without a latrine suggests **hookworm**; treat the cause (**albendazole 400 mg**) and the deficiency (**iron ~3 mg/kg/day for 3 months**), counsel on shoes and latrines, and recheck Hb at 4 weeks. A is wrong: she is compensated with Hb >4 g/dL; transfusion carries risk and is not indicated. B is wrong: iron without deworming leaves ongoing blood loss. C is wrong: praziquantel treats schistosomes, not hookworm, and there is no water-contact history. E is wrong: ivermectin is for Strongyloides; it does not replace iron and is not first line for hookworm.

2. B · Nil by mouth, nasogastric tube on free drainage, IV fluids with potassium, and urgent surgical review

B is correct: this is **partial small-bowel obstruction from an Ascaris bolus** without peritonism; start conservative management (**NBM, NGT decompression, IV fluids/electrolytes, analgesia**) with early surgical review; anthelmintic per the surgeon once the obstruction settles. A is wrong: oral treatment during obstruction may be vomited or provoke worm activity, and discharge is unsafe. C is wrong: surgery is for peritonitis, volvulus, intussusception or failure of conservative treatment. D is wrong: home treatment is unsafe for obstruction. E is wrong: barium is contraindicated in suspected obstruction and adds nothing to the ultrasound.

3. E · Give ivermectin 200 µg/kg (about 4 mg) and then start prednisolone

E is correct: diarrhoea, an itchy buttock rash (larva currens) and eosinophilia suggest **Strongyloides**, which routine microscopy often misses; steroids can trigger fatal **hyperinfection**, so give **ivermectin 200 µg/kg** (19 kg → about 4 mg) before or as steroids start. A is wrong: a single wet prep has very low sensitivity for Strongyloides. B is wrong: nephrotic syndrome needs prompt steroids; empirical treatment makes them safe. C is wrong: praziquantel does not treat Strongyloides. D is wrong: delaying treatment for repeated tests is unnecessary when empirical ivermectin is cheap and safe.

4. A · Sticky-tape test of the perianal skin first thing in the morning before washing

A is correct: nocturnal perianal itch in several family members suggests **pinworm (Enterobius)**, whose females lay eggs on the **perianal skin at night**; the **sticky-tape test** in the morning (repeat on 3 mornings) detects them. B is wrong: eggs are rarely present in stool. C is wrong: pinworm rarely causes eosinophilia (no tissue migration). D is wrong: urine is for S. haematobium. E is wrong: ultrasound cannot show pinworms.

5. C · Albendazole 400 mg for the child and family members (200 mg for any child aged 12–23 months), repeated after 2 weeks, plus nail-cutting, handwashing and washing bedding

C is correct: pinworm reinfects through fingers and bedding, so **treat the whole household** with **albendazole single dose, repeated after 2 weeks** (eggs hatch after the first dose), plus hygiene measures. A is wrong: praziquantel is for flukes and tapeworms. B is wrong: ivermectin is not first line and treating only the child leads to reinfection. D is wrong: the 3-day course is for heavy Trichuris; the key for pinworm is the repeat dose and family treatment. E is wrong: antihistamines relieve itch but do not kill the worms.

6. D · Oesophageal varices from presinusoidal portal hypertension due to Schistosoma mansoni

D is correct: a lakeside fisherman's son with **massive splenomegaly, left-lobe hepatomegaly, haematemesis and preserved liver function** (normal albumin and INR, no jaundice) has **hepatosplenic S. mansoni** with **periportal fibrosis → presinusoidal portal hypertension → varices**. A is wrong: there is no NSAID history, and it would not explain the huge spleen. B is wrong: cirrhosis would impair synthetic function, and alcohol is unlikely at 12. C is wrong: HMS causes a huge spleen but not varices and haematemesis. E is wrong: hepatitis B cirrhosis would usually show low albumin, raised INR or jaundice.

7. B · Praziquantel 40 mg/kg (1.2 g) single dose once stable

B is correct: **praziquantel 40 mg/kg** (30 kg → 1.2 g) kills adult schistosomes, stopping new egg deposition; early periportal fibrosis in children can regress. A is wrong: albendazole does not kill schistosomes. C is wrong: ivermectin is for Strongyloides. D is wrong: niclosamide treats tapeworms only. E is wrong: treatment prevents further damage and fibrosis may partly regress in children.

8. E · Microscopy of a filtered midday urine sample for eggs with a terminal spine

E is correct: **terminal haematuria** with river exposure suggests **S. haematobium**; eggs (terminal spine) are best found in **midday (10:00–14:00) urine** after filtration. A is wrong: no pyuria, nitrites or dysuria; UTI is unlikely. B is wrong: invasive and not indicated. C is wrong: stool Kato-Katz detects S. mansoni, not S. haematobium. D is wrong: post-streptococcal GN gives oedema, hypertension and brown urine with casts, which he does not have.

9. A · Albendazole 400 mg once daily for 3 days, iron, and gentle reduction of the prolapse

A is correct: barrel-shaped eggs with polar plugs are **Trichuris**; heavy infection causes **Trichuris dysentery syndrome, rectal prolapse, anaemia and clubbing**. Single-dose cure rates are poor, so give **albendazole 400 mg daily for 3 days**, iron, and reduce the prolapse (it usually resolves once the worms are cleared). B is wrong: a single dose often fails in heavy whipworm infection. C is wrong: the stool shows Trichuris eggs, not a bacterial dysentery picture. D is wrong: there is no evidence of amoebiasis. E is wrong: surgery is rarely needed; the prolapse resolves with treatment.

10. C · Brain imaging (CT or MRI) and referral, with anticonvulsants; treat the father's tapeworm with praziquantel

C is correct: focal seizures in a child whose household contact carries a **T. solium** tapeworm suggest **neurocysticercosis**; she needs **imaging** to define cyst number and stage, **anticonvulsants**, and specialist decisions about albendazole and steroids (UCG: refer). The **father must be treated** (praziquantel 5–10 mg/kg or niclosamide) to stop egg shedding. A is wrong: antiparasitic treatment without imaging can trigger inflammation, seizures and raised ICP. B is wrong: single-dose albendazole does not treat NCC and discharge ignores the seizures. D is wrong: there is no fever or meningism. E is wrong: she is afebrile, 9 years old and the seizures are focal.

11. D · Vitamin A 100,000 IU only; start deworming at 12 months

D is correct: routine **vitamin A** starts at **6 months** (100,000 IU for 6–11 months) while routine deworming starts at **12 months** (UCG 2023; WHO). A is wrong: no routine albendazole under 12 months, and 200,000 IU is the dose for ≥12 months. B is wrong: 400 mg albendazole is for ≥2 years and not for infants. C is wrong: no routine mebendazole under 1 year. E is wrong: praziquantel is given from 2 years and only in schistosomiasis-endemic programmes.

12. B · Albendazole 400 mg single dose plus praziquantel 40 mg/kg (560 mg) single dose with food, and iron

B is correct: polyparasitism is common at the lake; give **albendazole 400 mg** (≥2 years) for hookworm and **praziquantel 40 mg/kg** (14 × 40 = 560 mg) with food for S. mansoni, plus iron for the anaemia. A is wrong: ivermectin is avoided under 15 kg and does not treat schistosomes. C is wrong: 5 mg/kg praziquantel is the tapeworm dose, too low for schistosomiasis. D is wrong: 200 mg albendazole is for 12–23 months, and niclosamide treats tapeworm only. E is wrong: benzimidazoles do not kill schistosomes.

13. E · Löffler syndrome from larval migration through the lungs (hookworm or Ascaris), before eggs appear in the stool

E is correct: an itchy foot rash (larval skin penetration: 'ground itch'), then **cough, wheeze, migratory infiltrates and marked eosinophilia** with a **negative stool** (prepatent period) is **Löffler syndrome** from migrating larvae. Treat with an anthelmintic later (when adults are in the gut) and support symptoms. A is wrong: bacterial pneumonia does not cause marked eosinophilia or migratory shadows. B is wrong: a first episode of wheeze with eosinophilia and infiltrates after a skin rash fits larval migration better. C is wrong: TB has a subacute course with weight loss and contact history. D is wrong: no choking episode, and a foreign body gives fixed, not migratory, signs.

14. A · Biliary ascariasis

A is correct: **biliary ascariasis** presents with biliary colic, vomiting and mild jaundice; ultrasound shows a **worm (tube with a central line, 'railway track') in the bile duct**; worms may migrate after anthelmintics or fever. Manage conservatively (NBM, analgesia, antispasmodics, antibiotics if cholangitis), with ERCP if persistent. B is wrong: viral hepatitis lacks colic and a duct mass. C is wrong: pigment stones would show acoustic shadowing, not a tubular worm, and there is no SCD history. D is wrong: hydatid disease gives cystic liver lesions. E is wrong: appendicitis does not cause jaundice or duct dilatation.

15. C · It can persist for decades through autoinfection and cause fatal hyperinfection when steroids are given

C is correct: Strongyloides completes its life cycle within the host (**autoinfection**), so it persists for decades, and **immunosuppression (steroids, HTLV-1)** can trigger **hyperinfection** with Gram-negative sepsis and meningitis. A is wrong: larvae are few and intermittent; routine wet preps miss most infections (use concentration, agar plate or serology). B is wrong: WHO notes it is **not** sensitive to albendazole mass treatment; ivermectin is needed. D is wrong: it is acquired through skin penetration by soil larvae; pork transmits Taenia solium. E is wrong: terminal haematuria is S. haematobium.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Vomiting worms and a blocked bowel

You are the intern on the paediatric surgical ward at a regional referral hospital. A 3-year-old boy from Wakiso is admitted with 2 days of colicky abdominal pain, bilious vomiting and no stool, after vomiting a long pink worm at home. He has never been dewormed. Weight 11.5 kg, height 89 cm, MUAC 13.8 cm.

1. What are your first priorities?

> ETAT: airway (vomiting: positioning, suction), breathing, **circulation** (HR 140, CRT 2 s: mild dehydration, not shocked), glucose. **NBM**, **NGT on free drainage**, IV access, bloods (Hb, electrolytes, group and save), **IV fluids**: correct deficit (mild, 5% ≈ 575 mL over 24 h) plus maintenance 1075 mL/day, with **potassium** once passing urine. Analgesia.

2. What will you look for on abdominal examination, and which findings would mean surgery?

> Distension, visible peristalsis, **mobile sausage-shaped mass** (worm bolus), bowel sounds, tenderness. **Surgery** if **peritonism** (guarding, rebound), fever with tachycardia out of proportion, **blood per rectum** (intussusception/gangrene), signs of volvulus, free air on X-ray, or failure of conservative care in 24–48 h.

3. Ultrasound shows a 'whirlpool' bolus of worms in the jejunum; erect X-ray shows air–fluid levels; K+ 3.1 mmol/L. No peritonism. What is your plan?

> **Partial small-bowel obstruction from an Ascaris bolus**: conservative management for 24–48 h: NBM, NGT, IV fluids with **KCl**, 4-hourly abdominal review and girth measurement, strict input/output. **Anthelmintic per the surgeon** once obstruction settles (classically piperazine, which paralyses worms flaccidly, where available; then albendazole later). Avoid giving albendazole during complete obstruction.

4. On day 2 he passes many worms in stool, the distension settles and he is hungry. What next?

> Remove NGT, start oral fluids then feeds; give **albendazole 400 mg** (≥2 years) once tolerating oral intake (if not already given per surgeon), repeat routinely every 6 months. Check Hb and growth; vitamin A if due; **deworm the household**; WASH counselling (latrine, handwashing, nails, shoes, washing vegetables).

5. Why do children aged 1–5 get Ascaris obstruction more than adults?

> Their **small-bowel lumen is narrow**, they have **heavy worm loads** from frequent soil contact and hand-to-mouth behaviour, and irritation (fever, drugs, anaesthesia) causes worms to **tangle into a bolus** that blocks the narrow ileum or jejunum.

Take-away: An Ascaris bolus is a surgical problem first: decompress, rehydrate and watch for peritonism; deworm once the bowel recovers, then fix the household's sanitation.

Case 2: Pallor and a big spleen on the shores of Lake Victoria

At a HC IV in Mayuge district, a 9-year-old boy is brought by his aunt for tiredness, poor school performance and a 'swollen tummy'. He swims and fishes in the lake daily and walks barefoot. Weight 22 kg, height 121 cm. Last dewormed 14 months ago.

1. What will you look for on examination?

> Pallor (palms, conjunctivae), koilonychia, heart rate and signs of heart failure, **oedema**, jaundice; abdomen: **liver (left lobe), spleen size**, ascites, prominent veins; skin; growth: **Z-scores** (weight-for-age −2.1, height-for-age −2.3 = **stunted**); MUAC.

2. He is pale, HR 108, spleen 4 cm, left lobe of the liver palpable, no jaundice. HemoCue Hb 6.8 g/dL. Stool wet prep: hookworm eggs and eggs with a lateral spine. Interpret.

> **Polyparasitism**: **hookworm** (iron-deficiency anaemia) and **S. mansoni** (lateral-spine eggs) with early **hepatosplenic** involvement. Anaemia is moderate–severe but compensated (no heart failure; Hb >4).

3. Prescribe treatment with doses.

> **Albendazole 400 mg** single dose; **praziquantel 40 mg/kg = 880 mg** (≈ 1½ × 600 mg tablets) single dose with food (can split into two doses 4–6 h apart); **elemental iron ~3 mg/kg/day ≈ 66 mg/day** (e.g. ferrous sulphate 200 mg tablet once daily ≈ 65 mg elemental) with folic acid for **3 months**; malaria RDT (and treat if positive) before starting iron. No transfusion (Hb >4, compensated).

4. Which investigation would help assess his liver disease and what would you expect?

> **Abdominal ultrasound**: **periportal fibrosis** (thickened echogenic portal tracts, 'pipe-stem'), splenomegaly, portal vein diameter; LFTs/albumin usually **normal** (presinusoidal disease). FBC: microcytosis, **eosinophilia**. Refer if varices suspected (any haematemesis or melaena).

5. What prevention and follow-up will you arrange?

> Recheck Hb at **4 weeks** (expect +1–2 g/dL); repeat deworming every 6 months and **annual praziquantel** through the school programme; counsel on **shoes**, latrine use, avoiding lake water contact where possible, safe water; nutrition support for stunting; teach the aunt warning signs (vomiting blood, black stools, abdominal swelling).

> **Take-away:** Lakeside children often carry several worms at once: treat hookworm with albendazole plus iron and S. mansoni with praziquantel 40 mg/kg, and watch the spleen for portal hypertension.

Case 3: Steroids, diarrhoea and a hidden worm

A 6-year-old girl (19 kg) from Tororo is diagnosed with nephrotic syndrome at a regional referral hospital (heavy proteinuria, albumin 18 g/L, generalised oedema). The plan is prednisolone 60 mg/m2/day. She has had intermittent loose stools for months and an itchy rash around her buttocks that 'moves'.

1. Why should you think about worms before starting prednisolone?

> In an endemic area, **Strongyloides** may be present silently (autoinfection for years). High-dose steroids suppress cell-mediated immunity and can trigger **hyperinfection/dissemination** with Gram-negative sepsis and meningitis, often fatal. Her chronic loose stools and **moving rash (larva currens)** raise suspicion.

2. Her FBC shows eosinophils 1.4 x 10^9/L; a single stool wet prep is negative. What do you conclude and do?

> A single wet prep has **low sensitivity** for Strongyloides; eosinophilia plus symptoms make it likely. **Treat empirically: ivermectin 200 µg/kg ≈ 4 mg** (she weighs 19 kg, above the 15 kg threshold) once daily for 1–2 days, plus **albendazole 400 mg** for other STH. If possible, send concentrated stool/agar plate culture or serology. Start prednisolone without long delay.

3. Two weeks into prednisolone she develops fever, abdominal pain, cough, and drowsiness; blood culture grows E. coli. What do you suspect?

> **Strongyloides hyperinfection** (larvae carrying gut bacteria across the bowel wall → Gram-negative sepsis ± meningitis), or steroid-related sepsis/peritonitis (nephrotic children are also prone to pneumococcal peritonitis). Look for larvae in stool and sputum.

4. How do you manage suspected hyperinfection?

> ABC and sepsis care: IV access, fluids carefully (nephrotic), **ceftriaxone** (cover Gram-negatives; LP if safe for meningitis), **ivermectin 200 µg/kg daily until stool/sputum negative for 2 weeks**, **reduce steroids** to the lowest safe dose after discussion with the nephrologist, HDU/ICU referral, check HIV/HTLV-1 if available.

5. What is the key lesson for future steroid prescriptions?

> Before high-dose steroids (nephrotic syndrome, TB meningitis, severe asthma courses, leukaemia), **screen for or empirically treat Strongyloides** in children from endemic areas; it costs little and prevents a highly lethal complication.

> **Take-away:** Strongyloides hides from routine stool tests and from albendazole; give ivermectin before high-dose steroids in a child from an endemic area.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.